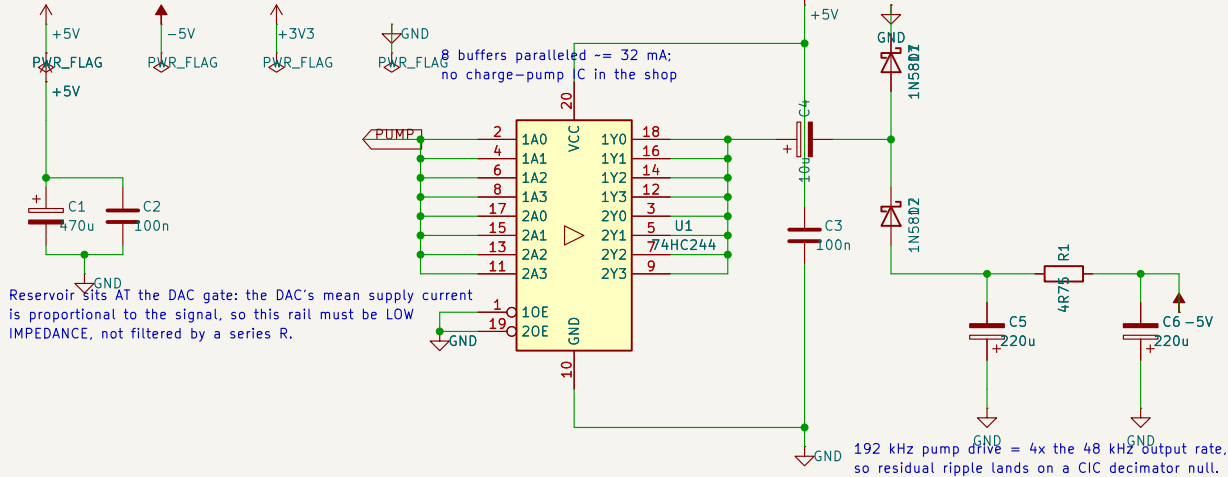


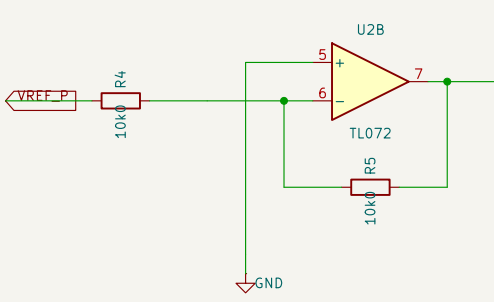
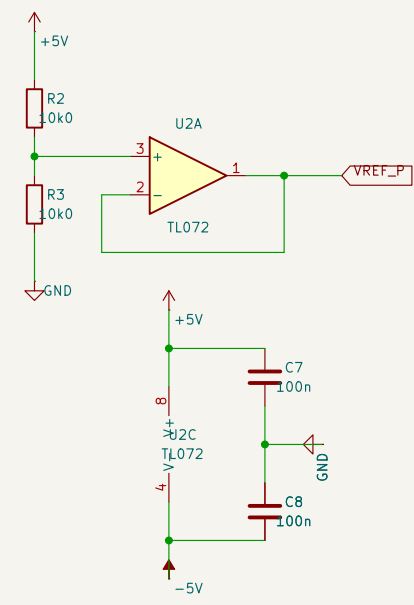
POWER (Pi +5V -> charge-pump -5V -> +/-2.5V DAC reference)



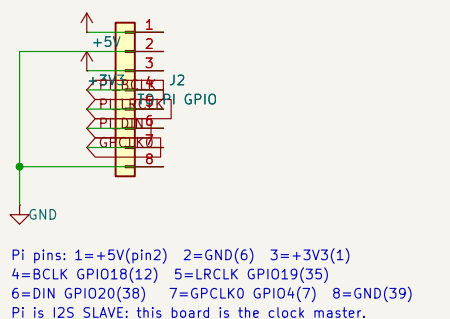
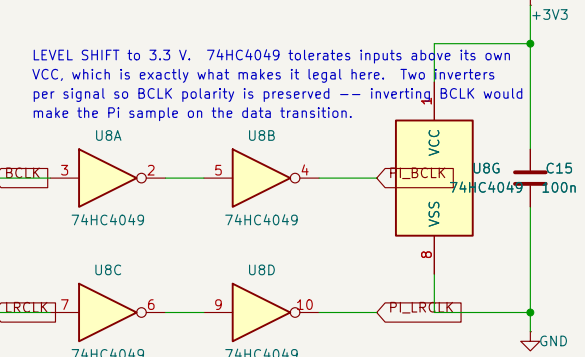
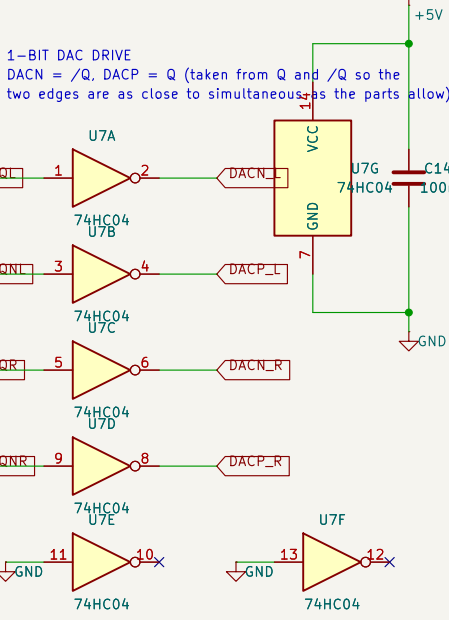
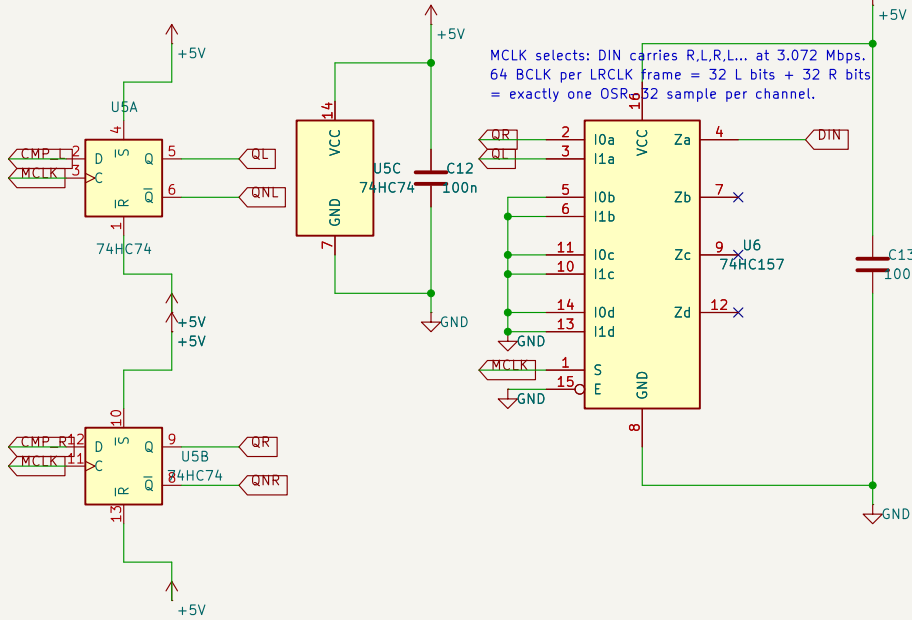
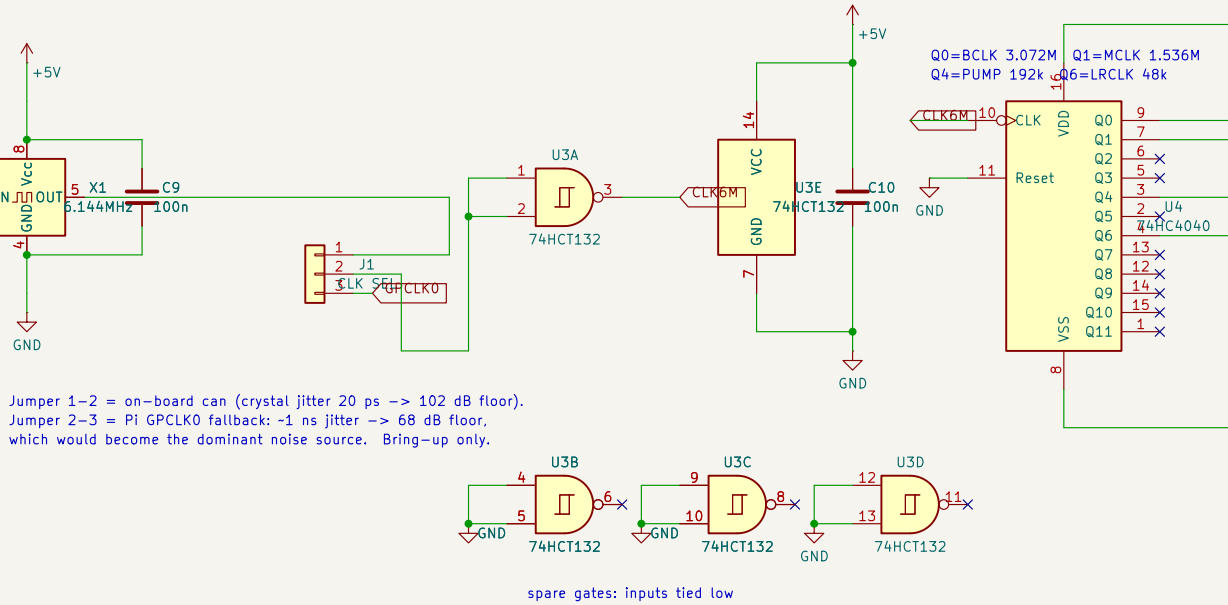
SPICE TESTBENCHES (sim/, one project each -- click to open)

- A rails and the -5 V charge pump
- B the +/-2.5 V ratiometric reference
- C clock buffer and 74HC4040 divider
- D integrator 1 with a real TL074
- E LM311 quantiser on a single supply
- F the 1-bit DAC and its offset leg
- G retiming, interleave and the 3.3 V level shift
- H one complete channel, with the click test

REFERENCE (+/-2.5V, ratiometric -- tracks +5V so rail noise cancels as gain, not additive noise)

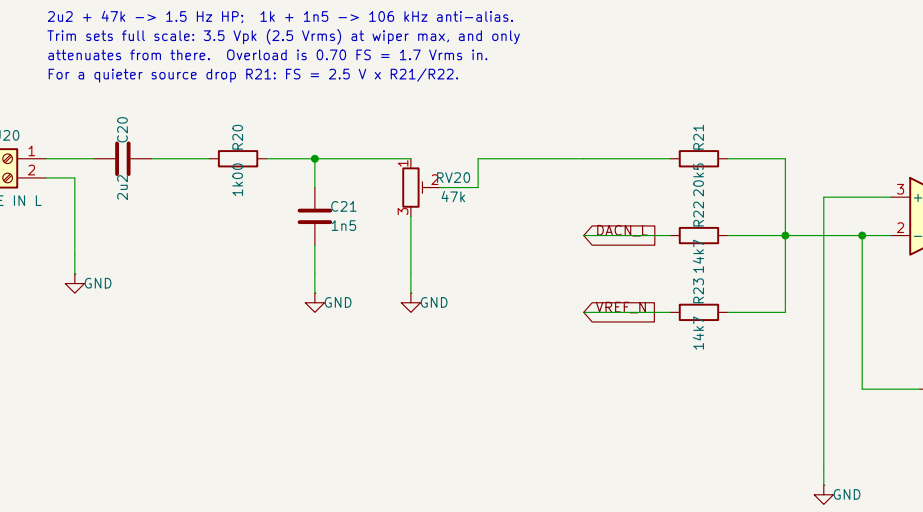


CLOCK AND DIGITAL (6.144 MHz -> /2 BCLK 3.072M, /4 MCLK 1.536M, /128 LRCLK 48k; L and R interleaved onto one DIN)

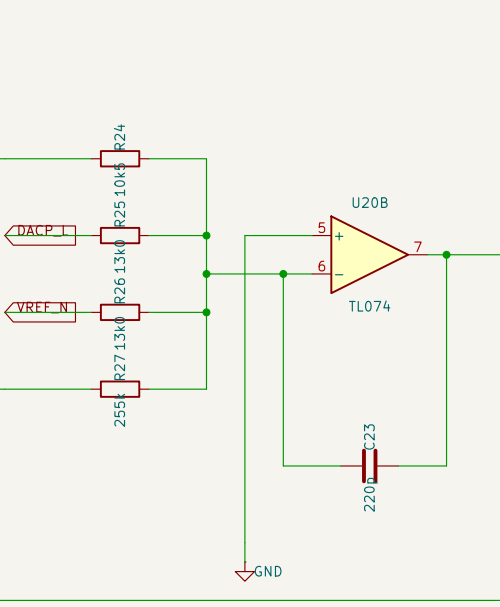


CHANNEL L - 3rd-order CT delta-sigma, fs 1.536 MHz, OSR 32

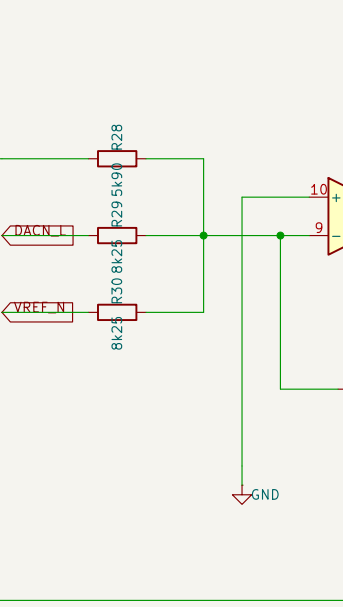
INTEGRATOR 1 (a1 = 0.247)



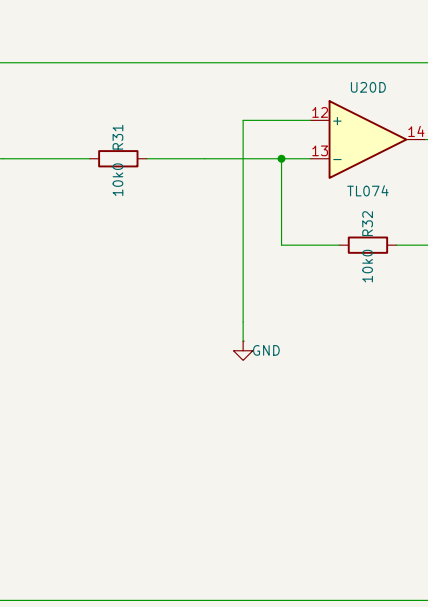
INTEGRATOR 2 (a2 = 0.321, + resonator)



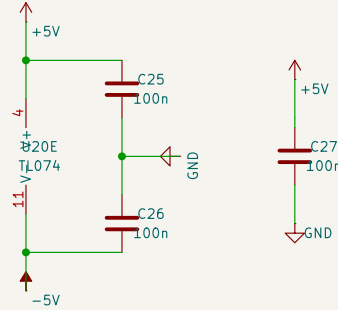
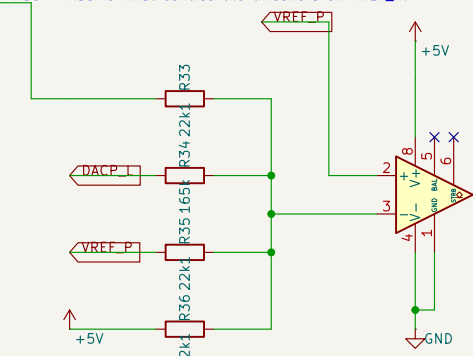
INTEGRATOR 3 (a3 = 0.611)



RESONATOR INVERTER (g = 0.0297)

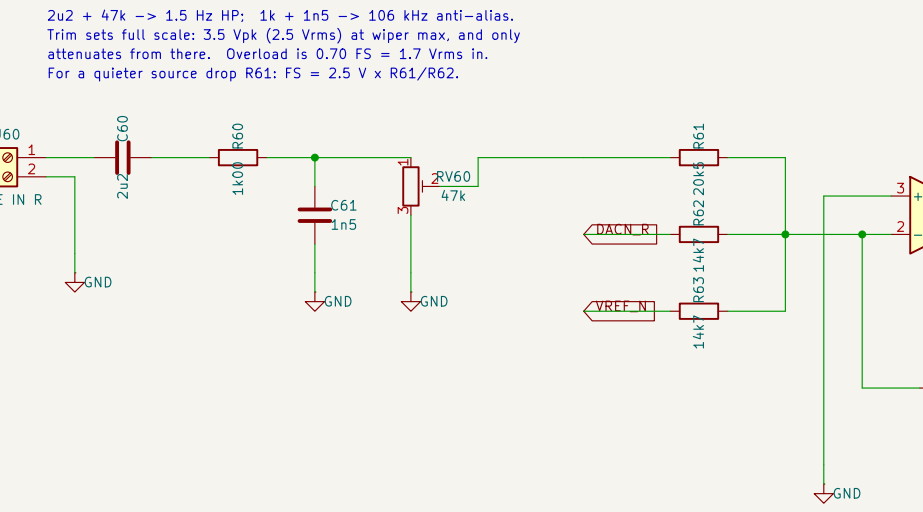


QUANTISER (LM311 + retiming D-FF)
R34 is the excess-loop-delay compensation: without it the LM311's 200 ns costs 8 dB and most of the overload margin. R36 = R33 is what centres the threshold on VREF_P.

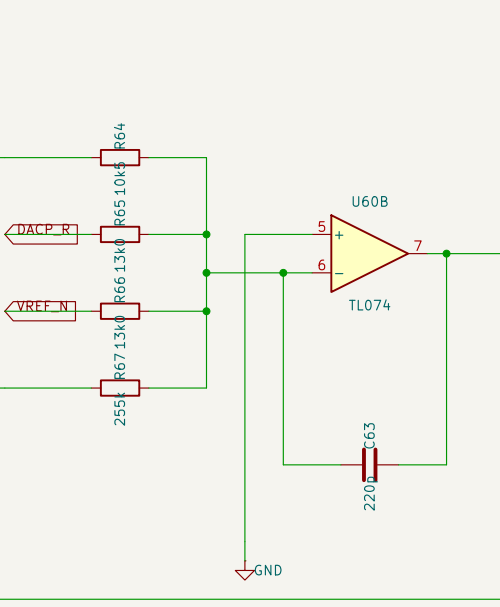


CHANNEL R - 3rd-order CT delta-sigma, fs 1.536 MHz, OSR 32

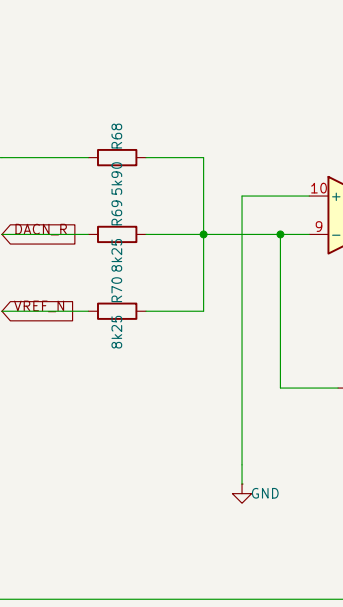
INTEGRATOR 1 (a1 = 0.247)



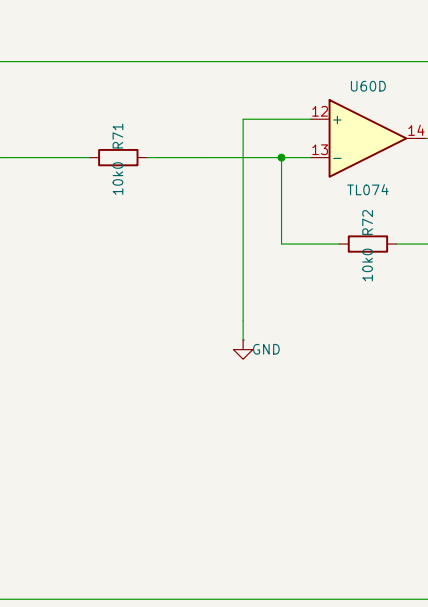
INTEGRATOR 2 (a2 = 0.321, + resonator)



INTEGRATOR 3 (a3 = 0.611)



RESONATOR INVERTER (g = 0.0297)



QUANTISER (LM311 + retiming D-FF)
R74 is the excess-loop-delay compensation: without it the LM311's 200 ns costs 8 dB and most of the overload margin. R76 = R73 is what centres the threshold on VREF_P.

